MILITARY SPECIFICATION

PRIMER, ELECTRIC, PA520

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements and quality assurance provisions for the fabrication, manufacture and packaging of electric primer, designated PA520, primarily designed for use in 30mm ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitations, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paper Board
PPP-B-601	-	Boxes, Wood, Cleated-Pleated
PPP-B-676	-	Boxes, Setup

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications for
MIL-A-82484	-	Adhesive and Sealing Compounds, Cellulose Nitrate Base, Solvent Type

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specifications to the extent specified herein:

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKING DRAWINGS

11825947	-	Cartridge, 30mm, Target Practice, M788
11825910	-	Primer, Electric

INSPECTION EQUIPMENT DRAWINGS

11825969	-	Test Barrel, 30mm Mann (6.5°)

PUBLICATIONS

US ARMY ARMY MATERIEL COMMAND

AMCR 715-505	-	Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges
ORD-SIP-S314	-	Visual Inspection Standard for Small Arms Ammunition Primer Defects

(Copies of specification, standards, drawings and publications required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following publications, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

Code of Federal Regulations

49CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1968 Edition and revision) available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20315. Orders for the above publication, cite 49CFR 71-90 (latest revisions).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Primer. The primer and all components thereof shall comply with Drawing (Dwg.) 11825910, all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article inspection. This specification contains technical provisions for first article inspections. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.4 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer and no deviation from that process shall be made without his prior approval. (see 6.10)

3.5 Insulation strength. The cup-insulator button assembly shall withstand an electrial potential of 900 volts direct current (DC) or 60 cycle alternating current (AC) for 0.20 second, minimum, without break-down of the insulator or flash over at the edges of the insulator.

3.6 Continuity and resistance. The primer circuit shall be continuous with an electrical resistance of the uncapped primer assembly (dry) not less than 20 nor more than 400 ohms. The electrical resistance of the capped primer shall not be less than 5 nor more than 400 ohms.

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3.7 Pellet weight. The total dry weight of the primer mix shall be as specified by Dwg. 11825910.

3.8 Firing voltage. The primer shall have an all-fire voltage of 22 volts DC or less and and a no-fire voltage of 3.0 volts DC minimum. The primer performance shall not be degraded when fired at 115 volts, 400Hz.

3.9 Electric primer time. The primer time shall not exceed .400 millisecond when energized by a 22 volt DC source with an 8.2 resistor in series with the primer.

3.10 Action time. The mean action time plus 3 standard deviations shall not exceed 4.0 milliseconds (ms) at -54°C, 21°C and 71°C. The maximum individual action time shall not exceed 4.0 ms (milliseconds) at -54°C, 21°C and 71°C. The action time is defined as the time period between the application of electric current to the primer and the exiting of the projectile from the muzzle.

3.11 Function and casualty. The primer shall function without casualty in the test cartridge.

3.12 Environmental conditions. After conditioning in accordance with 3.12.1, 3.12.2 and 3.12.3 the primer when assembled as a part of a complete round, as shown on Drawing 11825947, shall not deteriorate and shall meet the performance requirements of this specification.

3.12.1 Temperature, humidity and altitude. Temperature, humidity and altitude cycling condition with the temperature ranging from -54°C to 65°C, relative humidity up to 95 percent and pressure conditions equivalent to 15.2 kilometers altitude.

3.12.2 Aircraft vibration. Vibration as specified in MIL-STD-810, Method 514-2, Procedure IIb, except as modified in Section 4.

3.12.3 Extreme temperature storage. The primer, alone in Protective packages, or when assembled as part of a complete round shall be capable-of long term storage at the temperature extremes noted in paragraph 4.5.9.3.

3.13 Workmanship. The requirements for workmanship are as specified by applicable drawings, referenced specifications and the following:

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3.13.1 Processing defects. The primer shall be free of cracks, splits, deep scratches, dents, metal slivers, burrs, distortion, missing or unfinished component(s) and other forms of substandard workmanship.

3.13.2 Cleanliness. The primer shall be free of all foreign matter (see 6.9).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (See 4.3)
b. Quality Conformance Inspection (See 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Cup, Closure	11825912	125
Cup, Outer	11825913	125
Tapered Button, 22 volt, primer, Mod 1	11825911	125
Insulator	11825917	125
Cup-insulator button Assembly		125
Primer Assembly	11825910	1430

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4.3.2 Inspections to be performed. See MIL-A-48078 and Table therein.

4.3.2.1 First article inspection. Prior to submission of the first article sample to the Government, the contractor shall inspect all components and assemblies for all contract, drawings, and specifications requirements.

4.3.2.2 Examination and tests. The first article sample will be inspected at the Government laboratory or such facility specified in the contract. Determination as to approval of any first article sample shall be based only upon results of initial examinations and tests. (see 6.8)

4.3.3 Rejection. See MIL-A-48078 and the following:

4.3.3.1 First article sample failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63975(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Primer and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cup, Closure (DWG. 11825912) Examination for defects	125	100%	3.2	4.4.2.3
	Cup, Outer, (DWG. 11825913) Examination for defects	125	100%	3.2	4.4.2.1
	Tapered Button, 22 Volt Primer Mod I (DWG. 11825911) Examination for defects	125	100%	3.2	4.4.2.2
	Insulation (DWG. 11825917) Examination for defects	125	100%	3.2	4.3.2.2

NOTES:

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63975(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER See Below
	Primer and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Primer Assembly (DWG. 11825910)				
	Examination for defects	400	100%	3.2	4.3.2.2
	Electrical resistance (uncapped) 1/	100	Acc-0 Rej-1	3.6	4.5.2
	Pellet weight 2/	100	Acc-0 Rej-1	3.7	4.5.3
	Primer time 4/	100	Acc-0 Rej-1	3.9	4.5.5
	Firing voltage 3/	600	100%	3.8	4.5.4
	Insulation strength 7/	125	Acc-0 Rej-1	3.5	4.5.1
	Function & Casualty 6/	400	Acc-0 Rej-1	3.11	4.5.7
	Environmental conditioning			3.12	4.5.9
	Temperature-Humidity-Altitide	20		3.12.1	4.5.9.1
	Aircraft vibration	20		3.12.2	4.5.9.2
	Extreme temp.	40		3.12.3	4.5.9.3
	Action time 5/ -54°C	50		3.10	4.5.6
	21°C	50		3.10	4.5.6
	71°C	50		3.10	4.5.6

NOTES:

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1/ A primer which fails to comply with the applicable electrical resistance requirement shall be rejected.

2/ A primer which fails to comply with the applicable minimum dry weight requirement shall be rejected.

3/ Failure of any sample to meet the requirements of 3.8 shall be cause for rejection of the lot.

4/ If the sample average primer time plus three standard deviations exceeds .400 millisecond or if an individual primer time of the test sample exceeds .400 milliseconds, the lot shall be rejected.

5/ If the sample average action time plus four standard deviations exceeds 4 millisecond or if an individual primer time of the test sample exceeds 4 milliseconds, the lot shall be rejected.

6/ If a primer button is missing, a misfire occurs, or nine or more primers show evidence of leakage, as defined in AMCR 715-505, and ORD-SIP-S314 the lot shall be rejected. If more than three but less than nine primers show evidence of leakage, a second sample of double the number specified by Table I shall be tested. If in the accumulated samples, a button is missing, a misfire occurs, or nine or more primers show evidence of leakage, the lot shall be rejected.

7/ A cup-insulator-button assembly which fails to comply with the insulation strength requirement shall be rejected.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Each inspection lot shall contain:

a. Cartridge cases from one interfix lot number from one manufacturer.

b. Projectile parts from one interfix lot number from one manufacturer.

c. Primers from one lot number from one manufacturer.

d. Propellant from one lot number from one manufacturer.

e. Loaded cartridge case assemblies from one interfix lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63975 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825913
4.4.2.1	Cup, Outer				NEXT HIGHER ASSEMBLY 11825910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	See 4.4.2			
Major					
101	Crack, split or lamination		.25%	3.13.1	Visual
102	Dent, ragged edge (U or V)		.25%	3.13.1	Visual
103	Foreign matter, stain or corrosion		.25%	3.13.2	Visual
104	Wire-like edge, or slivers edged in or attached to cup		.25%	3.13.1	Visual
Minor					
201	Height		.40%	3.2	Gage
202	Outside diameter		.40%	3.2	Gage
203	Scratch or nick		.40%	3.2	Visual
204	Workmanship		.65%	3.13	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63975 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825911
4.4.2.2	Tapered Button, 22 volt, primer, Mod 1				NEXT HIGHER ASSEMBLY 11825910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	See 4.4.2			
Major					
101	Improperly formed		.25%	3.2	Visual
102	Crack, split or lamination		.25%	3.13.1	Visual
103	Dent		.25%	3.13.1	Visual
104	Foreign matter, stain or corrsion		.25%	3.13.2	Visual
105	Scratch or nick		.25%	3.13.1	Visual
Minor					
201	Overall height		.40%	3.2	Gage
202	Outside diameter		.40%	3.2	Gage
203	Workmanship		.65%	3.13	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11825912
4.4.2.3	Cup, Closure				NEXT HIGHER ASSEMBLY 11825910
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined	See 4.4.2			
Major					
101	Foreign matter, stain or corrosion		.25%	3.13.2	Visual
102	Ragged edges (U or V)		.25%	3.13.1	Visual
103	Flash hole missing or obstructed		.25%	3.13.1	Visual
104	Edged crooked or eared		.25%	3.2	Visual
105	Burr at flash hole		.25%	3.13.1	Visual
Minor					
201	Crack, split or lamination		.40%	3.13.1	Visual
202	Scratch, nick or dent		.40%	3.13.1	Visual
203	Height		.40%	3.2	Gage
204	Outside diameter		.40%	3.2	Gage
205	Workmanship		.65%	3.13	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63975 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.4	Primer, Assembly 1/		11825910
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Electrical resistance	All	100% 1/	3.6	
2.	Min. pellet weight	All	100% 1/	3.7	
3.	Missing, inverted or loss closure cup	See 4.4.2	100% 1/	3.2	Visual
4.	Oil or grease		100% 1/	3.13.2	Visual
5.	Inverted button		100% 1/	3.2	Visual
6.	Mixed types	All	100% 1/	3.2	Visual
7.	Insulation strength	All	100% 1/	3.5	4.5.1
8.	Firing Voltage	600	100%	3.8	4.5.4
Major					
101	Crushed or mutilated		.25%	3.13.1	Visual
102	Missing, misplaced or protruding insulator		.25%	3.2	Visual
103	Foreign matter other than oil or grease		.25%	3.13.1	Visual
104	No button or insulator covering button		.25%	3.2	Visual

NOTES:

1/ Inspection verification of prior 100% inspection shall be performed per (a) by utilization of contractor approved machine masters or (b) MIL-STD-105 AQL = .25, minimum sample, = 315 with zero defects allowed.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.4	TITLE Primer, Assembly 1/		SHEET 2 OF 2		DRAWING NUMBER 11825910
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
105	Presence of metal slivers on outside of primer or bridging the primer cup and button across insulator		.25%	3.13.1	Visual
Minor					
201	Missing, chipped or punctured disc		.40%	3.2	Visual
202	Cocked support cup		.40%	3.2	Visual
203	Explosive composition on disc or support cup		.40%	3.13.2	Visual 2/
204	Height		.40%	3.2	Gage
205	Diameter		.40%	3.2	Gage
206	Button depth		.40%	3.2	Gage
207	Workmanship		.65%	3.13	Visual

NOTES:

8/ Minute particles of explosive material are not considered cause for rejection.

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4.4.3 Inspection equipment. For the inspection equipment required to perform the examinations and tests prescribed herein, the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Insulation strength. The insulation strength of the cup-insulator-button assembly shall be determined applying the specified current, voltage and timing.

4.5.2 Electrical resistance. Each primer shall be tested for the electrical resistance limits of 3.6.

4.5.3 Pellet weight. The minimum total dry weight of the primer mixes shall be determined by measuring the recompressed bridge thickness of the primer. Bridge thickness is defined as the dimension between the center of the outside face of the primer button and the disc, or between the center of the outside face of the primer button and the exposed face of the primer support cup. The dry primer mixes in the primer assembly shall be recompressed prior to measuring the bridge thickness using force of approximately 100 pounds in-excess of the pressure required to move the cup support with respect to the primer cup. The minimum bridge thickness demension used to assure minimum total dry weight of the primer mix and the recompression force shall be established by the contractor and approved by the procuring activity.

4.5.4 Firing voltage. The method of test shall be as specified in AMCR 715-505 Volume 8 with the following exceptions:

a. Initial Voltage shall be 12 volts.
b. Voltage increments (decrements) shall be 2 volts.
c. Primers shall be fired at each voltage increment (decrement).
d. An 8.2 ohm resister shall be placed on series with the primer.
e. The primer firing device shall be set for a 1 millisecond pulse time.

If the average critical voltage plus 3 standard deviations exceeds 22 volts DC the lot shall be rejected. If the average critical voltage minus 3 standard deviations is less than 3 volts DC the lot shall be rejected.

4.5.5 Electric primer time. Test method and equipment shall be approved by the procuring agency.

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4.5.6 Action time. The test rounds, assembled in accordance with Drawing 11825947, shall be conditioned at -54°C, 21°C and 71°C and fired from a test fixture barrel in accordance with Drawing 11825969. The time between application of voltage to primer and exit of the projectile from the muzzle shall be measured. The time shall comply with the requirement of 3.10. Any corrected action time of 2.40 milliseconds or less shall be regarded as an instrumentation error.

4.5.7 Function and casualty. The method of test shall be as specified in AMCR 715-505, Volume 8. The test sample primers shall be inserted into cases that have been previously inspected for depth and diameter of primer pocket and length to shoulder basic diameter. The sample primed cases shall be assembled as Cartridge, 30mm, Target Practice, conforming to Drawing 11825947. Prior to firing, the test cartridges shall be inspected for depth of primer seating, primer crimp missing, loose primer, metal slivers on case head or primer, profile and alignment max., and presence of foreign matter on the primer button. The sample test cartridges shall be fired in bursts of 25 rounds (21°C), -54°C, and 71°C at a cyclic rate of the ADEN automatic gun, or the XM230E1 autogun as required in the contract or purcahse order. The gun barrel shall be at ambient temperature at the beginning of each burst. Two hundred cartridges shall be fired at 21°C, and 100 cartridges at each -54°C and 71°C.

4.5.8 Misfires. Test cartridge(s) which fail to fire using the specified equipment shall be classified as to the cause of the failure where applicable and recorded as misfire(s) in accordance with Table II.

4.5.9 Environmental tests. Unless otherwise specifed, performance during and/or after environmental exposure, as specified in the applicable test procedure shall consist of conducting the test of 4.5.6 except all test rounds shall be conditioned at 21° C prior to firing.

4.5.9.1 Temperature, humidity, and altitude. The test cartridge shall be subjected to the temperature/humidity/altitude test of MIL-STD-810, method 518. After exposure, the test item shall be examined for deterioration and subjected to the test referenced in 4.5.6.

4.5.9.2 Aircraft vibration. The test cartridge(s) shall be subjected to a broad band random vibration profile in accordance with the requirements of MIL-STD-810, Method 514.2, Procedure IIb, except the vibration spectrum shall be as shown in Figure 1. The test item shall be mounted to a test fixture simulating a AAH magazine in a normal airplane mounting specified by the procuring agency. The test duration shall be 45 minutes in each of three mutually perpendicular axs at each of two temperature conditions, -40° + 2.8°C and 65.5° + 2.8°C. Following exposure, all of the test items shall be subjected to the tests referenced in 4.5.6.

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4.5.9.3 Extreme temperature storage. 20 samples shall be subjected to test of MIL-STD-810, Method 502.1 and the 20 remaining samples shall be subjected to the test of MIL-STD-801, Method 501.1. After reconditioning in accordance with 4.5.9, the samples shall be fired in accordance with 4.5.6.

4.5.10 When cartridges are required for acceptance tests cartridges shall contain the following:

a. Cartridge cases from one interfix lot number from one manufacturer.

b. Projectile parts from one interfix lot number from one manufacturer.

c. Primers from one lot number from one manufacturer.

d. Propellant from one lot number from one manufacturer.

e. Loaded cartridge case assemblies from one interfix lot number from one manufacturer.

4.5.11 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms on the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

5. PACKAGING

5.1 Packaging and packing. Where packing is required, the primers shall be packed in standard commercial containers so as to assure acceptance by common or other carrier for safe transportation, in accordance with Title 49, CFR 71-90 (ICC Rules and Regulations for the Transpiration of Explosives and Other Dangerous Articles), at the lowest rate, to the point of delivery. The primers shall be packed and spaced in such a manner that a barrier of appropriate material is present between the crown of one primer and the base of another primer to resist mass detonation and shield the primers from contact with stray electrical current, flame and shock. The primers shall be packed in such a manner as to be protected from moisture, having an outer container of weather resistant type fiberboard and all container seams and joints sealed with waterproof tape. The inner package shall be of water proof material.

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5.2 Marking. Where packing of primers is required, the packing boxes shall be marked as specified by MIL-STD-129. Marking shall idenfify the supplier, contract number, item nomenclature, lot number, drawing number and revision, primer mix number and month and year of manufacture.

5.3 Shipping. When shipments of more than one lot are made at one time, such as in a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6 . NOTES

6.1 Intended use. This specification covers the metal parts, assembling and packing for electric primer, designated PA520, for use in 30mm ammunition.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078 (AR). Submit equipment designs as required to Commander, ARRADCOM, ATTN: DRDAR-QAF-I, Dover, NJ 07801. Request letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.4 Submission of results of contractor-conducted examinations and test. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commander, ARRADCOM, ATTN: DRDAR-QAF-S.

6.5 Submission of test data. In addition to the normal distribution of records, when the cartridges are procured by ARRADCOM, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commander, ARRADCOM, ATTN: DRDAR-QAF-S, Dover, NJ 07801.

6.6 Standard deviation shall be calculated from the following formula.

$$s = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n}}$$

where: x_i = each individual value

$\bar{x}$ = sample arithmetic mean $\frac{\Sigma x_i}{n}$

n = sample size

$\Sigma (x_i - \bar{x})^2$ = the sum of the squares of the differences between the sample mean and each individual value.

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6.6.1 For a faster and easier method of arriving at the standard deviation formula, the following alternate method may be used:

$$= n \sqrt{\frac{(\Sigma x^2) - (\Sigma x)^2}{n\ (n-1)}}$$

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.8 First article. When a first article inspection is required, the contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, test and approval of the documents for first article.

6.9 Contamination. Particular care should be exercised to avoid contamination of primers by oil, grease and other foreign material since each contamination can increase action time and make the primer unsuitable for use in ammunition.

6.10 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

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